

The Alan Machine

João Alisson and contributors

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Preface

For Alan Turing, who imagined the machine,
and for Alan Ricardo.

How to read this book

This book is about a machine that does not exist. We call it Alan, and we use it to ask a question physics takes very seriously: can entropy be reversed, and if it could, what would happen to time?

You do not need a physics degree to read it. The chapters are written for anyone who is curious. When the math helps, it shows up in boxes marked *Going deeper*, which you can open or skip. The story works either way.

Some chapters have a companion paper, written for specialists, with the full argument. You will find the papers near the end of the book, and each chapter tells you when one exists.

Each part ends with an *interlude*: a short pause for a question that physics can inform but not answer, such as whether time really flows or whether a machine can think. After the main text, *Building Alan* looks at real technology, from today's chips to quantum computers, and asks how far each one is from the limits the chapters describe. Those pages are updated every year.

Along the way, we keep five kinds of statement apart. Unmarked claims are **established** physics, with references you can check. A claim that rests on a newer source, such as a preprint or a manufacturer's figure, says so in words: it is **reported**, not yet established. Boxes marked **Extrapolation** follow from established physics, but only if their stated assumptions hold. Boxes marked **Speculation** hold ideas that are conceivable but that no evidence supports yet. Boxes marked **Philosophy** hold questions that physics cannot settle, with the best arguments on each side. A thought experiment is only useful if you always know which of the five you are reading.

Every technical term links to its plain-language definition in the Concepts appendix at the end.

Part I

The Machine and the Question

1 Meet Alan

This chapter has not been written yet.

2 What Entropy Is

This chapter has not been written yet.

3 The Arrow of Time

This chapter has not been written yet.

Interlude: Does Time Flow?

This interlude has not been written yet.

Part II

Information Is Physical

4 Maxwell's Demon and the Szilard Engine

This chapter has not been written yet.

5 Landauer's Principle

This chapter has not been written yet.

6 Reversible Computing

This chapter has not been written yet.

7 Can a Mind Beat the Second Law?

This chapter has not been written yet.

Interlude: Three Demons

This interlude has not been written yet.

Part III

The Limits of the Machine

8 How Fast Can It Compute

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9 How Much Can It Hold

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10 Is Alan a Quantum Computer?

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Interlude: Are We Inside an Alan?

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Part IV

Reversing Entropy

13 Local Reversal

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14 Life: The Original Local Reversal

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15 Fluctuations and Recurrence

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Part V

The Impact on Time

17 What Time Is

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18 Would Time Run Backward

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19 Closed Timelike Curves

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20 The End of the Universe

This chapter has not been written yet.

Interlude: Can Alan Think?

This interlude has not been written yet.

Epilogue: What Alan Cannot Do

This chapter has not been written yet.

Part VI

Building Alan

About Building Alan

The chapters ask what physics allows. This part asks a different question: how close does today's technology come, and what would close the gap? The answer changes every year, so these pages are living documents. Each one has a review date, and every number in it carries the date it refers to and its source.

Each living review looks at one technology or one bottleneck and measures it against the same yardsticks:

- **The Landauer bound:** the least heat that erasing one bit can release, $k_B T \ln 2$ at the temperature the device runs at (Landauer 1961; Bérut et al. 2012).
- **The Margolus-Levitin bound:** the most operations per second that a given amount of energy allows (Margolus and Levitin 1998; Lloyd 2000).
- **The Bekenstein bound:** the most information that a given mass and size can hold (Bekenstein 1981).
- **Energy at the wall plug:** what the whole system draws from the grid, cooling and power delivery included. A device that runs near the Landauer bound at 4 K can still cost more, counted at the wall plug, than one at room temperature.

These pages are written for engineers and researchers who design processors, memory, AI hardware and data centers, and for any reader who wants to know where real machines stand. The numbers behind them are published as data (see [the data folder](#)), so they can be reused and checked.

The same method, applied beyond computing, is [Escape Velocity](#): an open atlas of what each technology still needs to reach maturity, from fusion to medicine, and of which technologies wait on which. Its entries on quantum computers, data movement, heat removal and the energy of AI inference link back to these living reviews.

Living reviews

Living review	Status	Last reviewed	Review due	Curators
CMOS: The Baseline	proposed	never	not reviewed yet	none yet

Living review	Status	Last reviewed	Review due	Curators
The Cost of Moving Data	proposed	never	not reviewed yet	none yet
Getting the Heat Out	proposed	never	not reviewed yet	none yet
Adiabatic and Reversible Logic	proposed	never	not reviewed yet	none yet
Superconducting Logic	proposed	never	not reviewed yet	none yet
Quantum Hardware	proposed	never	not reviewed yet	none yet
Thermodynamic and Probabilistic Computing	proposed	never	not reviewed yet	none yet
Neuromorphic Computing	proposed	never	not reviewed yet	none yet
Optical Computing	proposed	never	not reviewed yet	none yet
The Energy of a Token	proposed	never	not reviewed yet	none yet

CMOS: The Baseline

This living review has not been written yet.

The Cost of Moving Data

This living review has not been written yet.

Getting the Heat Out

This living review has not been written yet.

Adiabatic and Reversible Logic

This living review has not been written yet.

Superconducting Logic

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Quantum Hardware

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Thermodynamic and Probabilistic Computing

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Neuromorphic Computing

This living review has not been written yet.

Optical Computing

This living review has not been written yet.

The Energy of a Token

This living review has not been written yet.

Part VII

Papers

About the Papers

The chapters tell the story. The papers make the argument.

Some chapters have a companion paper with the full scientific treatment of their subject: the derivations, the assumptions and the references a specialist would expect. Each paper stands on its own, with its own authors, abstract and references, and can be cited independently of the book. A chapter can lean on several papers, and a paper can support several chapters.

No papers have been written yet.

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Contributing

This book is written in the open, and anyone can improve it. You do not need to know git, and you do not need to be a physicist: many of the most useful contributions are a single error found by a careful reader.

What you can contribute

- **An error:** a typo, a broken link, or a wrong equation or number.
- **A challenge:** a claim you think is wrong, unsupported, or classified as the wrong kind. The book marks every claim as established, reported, an extrapolation, speculation or philosophy, and a wrong label is an error too.
- **A reference:** a paper or book that should be cited, or listed as further reading.
- **New numbers** for a Building Alan living review, with their unit, date and source.
- **New text:** a chapter, a section, an interlude, a living review or a paper, proposed before it is written.
- **A translation**, or a correction to one.
- **An idea** that is not a proposal yet (“what if Alan could...”).

Without git: a form on GitHub

You need only a free GitHub account. Each form asks for what a reviewer needs; write in English.

Form	Use it for
Erratum	A typo, a broken link, or an error in an equation or a number
Scientific challenge	A claim that is wrong, unsupported, or classified as the wrong kind
Content proposal	A new chapter, section, interlude, living review or paper
Living review update	New or corrected numbers for a Building Alan living review

Form	Use it for
Reference suggestion	A paper or book that should be cited or listed as further reading
Translation	A wrong or missing translation, or an offer to translate

Ideas and questions go to [Discussions](#). The people who look after the part of the book you name are told about your issue, and they answer it there.

With a pull request

To change the text yourself:

1. Open “Edit this page” at the foot of a page for a small fix in the browser, or fork the [repository](#) and work on your own copy.
2. Make the change, keeping the book’s [style guide](#).
3. Open a pull request and fill in its template.
4. A person who looks after those pages reviews it, and a maintainer merges it.

The title of the pull request, the sign-off every commit needs, the checks to run and the step-by-step setup are in [CONTRIBUTING.md](#), which has the complete rules. Translators start from [docs/translating.md](#).

Who looks after the book

Every part of the book has moderators, and pages can have curators who keep them accurate. They are listed in Appendix [E](#), with the maintainers of each translation. How to take one of these roles is in [GOVERNANCE.md](#).

A Concepts

Every technical term the book uses, in plain language. Each entry lists the pages that use it.

Arrow of time

The difference between past and future that we observe in everyday life: eggs break but do not unbreak, and we remember yesterday but not tomorrow. It stands out because the fundamental laws of physics barely distinguish between the two directions of time.

Related: [Entropy](#), [Second law of thermodynamics](#)

Bit

The word has two meanings that are kept apart. A bit is a binary digit: one place in a memory or a message that holds 0 or 1, so ten such places hold ten bits. It is also the everyday name of the unit of information, whose own name is the shannon (symbol Sh). Learning which of two equally likely answers is the true one gives one shannon; an answer that could be predicted gives less, and a certain one gives none. So ten binary digits carry at most ten shannons, and exactly ten only when every sequence is equally likely. Claude Shannon set out the measure in 1948: the logarithm to base 2 of the number of equally likely possibilities. Taken with the natural logarithm instead, the same information is measured in nats, and one shannon is $\ln 2$ nat, about 0.693 nat. The shannon is the unit named in the standard IEC 80000-13:2025. Entropy in thermodynamics is k_B times the information in nats, which is why Landauer's bound for erasing one bit, $k_B T \ln 2$, carries the factor $\ln 2$: it converts one shannon into nats.

Related: [Landauer's principle](#), [Entropy](#)

References: (Shannon 1948)

Closed timelike curve

A path through spacetime that returns to its own past. Such curves are not known to exist, but studying them has taught something about general relativity and quantum information. Following David Deutsch, a closed timelike curve is treated as a region of spacetime where nature must produce a consistent outcome: a fixed point of whatever process runs around the loop. Aaronson and Watrous showed in 2008 that, if such curves existed, quantum computers would be no more powerful than classical ones: both could solve exactly the problems that need a polynomial amount of memory (PSPACE).

Related: [Arrow of time](#)

References: (Deutsch 1991; Aaronson and Watrous 2008)

Entropy

A measure of how many microscopic arrangements of a system (its microstates) are compatible with what we observe at large scale (its macrostate). The more ways the parts can be arranged without changing how the whole looks, the higher the entropy. It grows with the logarithm of that number: the entropy is k_B times its natural logarithm, so doubling the number of arrangements adds $k_B \ln 2$, however large the system. Thermodynamics measures the same quantity through heat: heat taken in reversibly at temperature T , divided by T , is the change in entropy.

Related: [Second law of thermodynamics](#), [Arrow of time](#), [Bit](#)

Landauer's principle

Erasing one bit of information in surroundings at temperature T releases at least $k_B T \ln 2$ of heat. Forgetting has a minimum physical cost. The bound was proposed by Rolf Landauer in 1961 and confirmed experimentally in 2012.

Related: [Bit](#), [Reversible computing](#), [Maxwell's demon](#)

References: (Landauer 1961; Bérut et al. 2012)

Maxwell's demon

A thought experiment devised by James Clerk Maxwell in 1867: a tiny being opens and closes a door between two boxes of gas, letting fast molecules go one way and slow ones the other. It seems to lower entropy for free. The accepted resolution is that the demon must record what it sees, and erasing that record eventually pays the entropy back.

Related: [Second law of thermodynamics](#), [Landauer's principle](#)

Poincaré recurrence

A theorem of Henri Poincaré: any finite system comes arbitrarily close to its initial state after some very long but finite time. Taken statistically it is not a paradox, but it seems to violate the second law of thermodynamics, since the entropy of a gas would in time return near its starting value. The theorem leaves gravity out; with gravity, colliding particles can form black holes, which can quench the recurrence or change how long it takes.

Related: [Second law of thermodynamics](#), [Entropy](#), [Arrow of time](#)

References: (Dong and Stojkovic 2016)

Quantum speed limit

The shortest time in which a quantum system can evolve into a state fully distinguishable from the one it started in. Leonid Mandelstam and Igor Tamm bounded it by the spread of the system's energy; Norman Margolus and Lev Levitin, in 1998, by its average energy above the lowest possible: the time is at least $\pi\hbar/2E$, with E measured from the ground state. Each step of a computation has to reach a distinguishable state, so the limit caps how many steps per second a computer with a given energy can take.

Related: [Bit](#), [Reversible computing](#)

References: (Mandelstam and Tamm 1991; Margolus and Levitin 1998; Lloyd 2000)

Reversible computing

Computing without erasing information, so that every step can be run backward. Two meanings are kept apart. A computation is logically reversible when each state has only one possible predecessor, so the input can be recovered from the output; Charles Bennett showed in 1973 that a general-purpose computer can be made logically reversible at every step. It is thermodynamically reversible when it produces no entropy, a limit a real device approaches only

by running ever more slowly. Logical reversibility removes the minimum cost that Landauer's principle sets on erasure; it does not by itself remove the rest of the dissipation.

Related: [Landauer's principle](#), [Bit](#)

References: (Bennett 1973)

Second law of thermodynamics

In an isolated system, entropy does not decrease over time. It is a statistical law: a decrease is not forbidden, only overwhelmingly unlikely for systems with many parts.

Related: [Entropy](#), [Arrow of time](#), [Maxwell's demon](#)

Szilard engine

The simplest case of Maxwell's demon, studied by Leo Szilard in 1929: a box holding a single molecule, in thermal contact with a heat bath, and a partition that can be inserted and can slide along the box. With the partition inserted, the molecule pushes on it; if it moves the way it is pushed, it can lift a weight, and the most work this extracts is $k_B T \ln 2$, drawn from the heat bath alone. Szilard argued that the second law is saved if the demon's acquisition of knowledge, which side the molecule is on, comes with a compensating entropy cost.

Related: [Maxwell's demon](#), [Landauer's principle](#), [Second law of thermodynamics](#)

References: (Maroney 2009)

Turing machine

An abstract model of computation introduced by Alan Turing in 1936: a machine that reads and writes symbols on an unlimited tape, following a finite table of rules. Anything a modern computer can compute, a Turing machine can compute too, given enough time and tape.

Related: [Bit](#)

References: (Turing 1937)

B Notation

Every symbol keeps the same meaning across the whole book.

Symbol	Meaning	Unit
k_B	Boltzmann constant	J K^{-1}
T	Thermodynamic temperature	K
S	Entropy	J K^{-1}
E	Energy	J
$\hbar$	Reduced Planck constant, $\hbar/2\pi$	J s
c	Speed of light in vacuum	m s^{-1}

C Constants

Values from the 2019 revision of the SI, where the constant is exact by definition, and from CODATA otherwise.

Constant	Symbol	Value	Unit	Standard uncertainty	Source
Speed of light in vacuum	c	299792458	m s^{-1}	exact	SI (2019), exact by definition
Planck constant	h	$6.62607015 \times 10^{-34}$	J s	exact	SI (2019), exact by definition
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$	J s	exact	$h / 2 \pi$, exact; value truncated to 10 significant digits
Boltzmann constant	k_B	1.380649×10^{-23}	J K^{-1}	exact	SI (2019), exact by definition
Newtonian constant of gravitation	G	6.67430×10^{-11}	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$	1.5×10^{-15}	CODATA 2018

D A Short Primer on Statistical Mechanics

This appendix has not been written yet.

E Moderators and Curators

The people who look after the book. Maintainers look after the whole repository. Moderators look after a part: they review the changes to its pages and triage its issues. Curators look after single pages and review their living reviews. Translators look after the book in another language. People are named by their GitHub handle. How the roles work, and how to take one, is in [GOVERNANCE.md](#).

Maintainers

[JoaoAlisson](#)

Moderators

Part	Moderators
The Machine and the Question	none yet
Information Is Physical	none yet
The Limits of the Machine	none yet
Reversing Entropy	none yet
The Impact on Time	none yet
Building Alan	none yet
Papers	none yet

Curators

No page has a curator yet.

Translators

The maintainers of a language approve the translations into it; the reviewers have reviewed some of them. How to translate is in [docs/translating.md](#).

Language	Maintainers	Reviewers
Portuguese (Brazil)	JoaoAlisson	none yet